

Academic Integrity Pledge

EF234405 Design & Analysis of Algorithms — Final Exam (Group Capstone Project)

Project title:

Smart Delivery Routing: Dijkstra vs. Bellman-Ford on a Synthetic Road-Network Graph

Full name: _____

Student ID: _____

Class (D / IUP / E / F / G): _____

By signing below, I (the undersigned, the sole member of this team) affirm the following:

- This project is my own original work, produced specifically for this Final Exam. It has not been copied, in whole or in part, from the Appendix A specimen, from any other team's submission, from a previous offering of this course, or from any other unattributed source.
- The core algorithmic logic of both implemented algorithms (Dijkstra's algorithm and the Bellman-Ford algorithm, in `src/dijkstra.py` and `src/bellman_ford.py`) was written by me. External libraries (numpy, scipy, matplotlib) were used only in supporting roles — random number generation, nearest-neighbor graph construction, plotting, and timing — and not to call any pre-built shortest-path or graph algorithm.
- All reported experimental results (`benchmark_results.csv`, the runtime-vs-size plot, and the growth-exponent figures quoted in the report) were produced by actually running the code in this repository, with the fixed random seeds stated in the report and README, and have not been fabricated, cherry-picked, or altered.
- The GitHub commit history accurately reflects my own individual work on this project and has not been altered, backdated, or fabricated to misrepresent the timeline or authorship of the work.
- This project is submitted as a single-member team. The 100% contribution attributed to me in the Report's contribution table (§4.2) accurately reflects that I alone designed, implemented, tested, benchmarked, analysed, and wrote the report for this project.
- I understand that any confirmed violation of academic integrity in connection with this submission — including plagiarism, fabricated results, or a misrepresented commit history — is handled separately by the instructor and overrides the BC (61) floor grade described in the Final Exam specification.

Signature: _____

Date: _____